Shambhavi Kumari
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Internship

QC Microbiology Intern | Centrient Pharmaceutical Company — Jun 2025-Jul 2025

- Conducted media preparation and sterilization (MCB, SCDM, Peptone) and operated autoclaves under aseptic conditions, ensuring contamination-free testing.
- Performed water testing, product testing, and environmental monitoring (air, surface, and personnel sampling), following SOPs, GMP, and biosafety protocols.
- Carried out microbial culture inoculations and sterility testing using *Bacillus subtilis* and other strains, supporting pharmaceutical product quality control.
- Worked in Laminar Air Flow (LAF) benches for aseptic handling and maintained detailed SOP logbooks for media, reagents, and QC activities.
- Gained cross-functional exposure to the Bulk Drug Lab, Raw Material Lab, and Instrumentation Lab, learning analytical techniques including normality checks, moisture analysis, optical rotation, and chromatography methods.
- Strengthened knowledge of quality culture, GMP compliance, and pharmaceutical documentation practices, ensuring reliability in QC processes.

Projects

Cutter Mill with Sieve Separator | Mechanical design, prototype development, problem solving — Nov 2025

- Designed and built a prototype cutter mill integrated with a sieve separator for particle size reduction and separation.
- Developed a hopper-based feeding and grinding system powered by a motor.
- Implemented a sieve unit for particle size classification after milling.
- Demonstrated principles of size reduction and material processing used in pharmaceutical and biotech industries.

Heart Disease Prediction Using Machine Learning |Machine Learning, Data Analysis — Jul 2024

- Developed a machine learning model to predict the risk of heart disease using patient clinical data.
- Performed data preprocessing, feature selection, and dataset cleaning to improve model accuracy.
- Applied machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest for disease prediction.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix analysis.

Automated Washroom / Traffic Light Control System |8051 Microcontroller, Assembly Language — Aug 2023

- Designed and implemented an automated traffic light control system using 8051 microcontroller architecture to manage signal sequencing.
- Programmed the system in Assembly language to perform arithmetic, logical, and control operations for accurate signal timing.
- Interfaced LEDs, switches, timers, and seven-segment displays with the microcontroller to simulate real-world traffic flow and pedestrian signals.
- Implemented delay subroutines and timer interrupts to ensure precise control of signal timing and smooth transitions..

Certificates / Certifications

SQL Beginner to Advanced – Code Basics — **Aug 2025**

Business English – Saylor Academy — **Jun 2024**

Skill Development Course – Lovely Professional University — **Sep 2023**

Fundamental English MOOC – Saylor Academy — **Sep 2023**

Skills

Languages: Python, BioPython

Soft Skills: Communication, teamwork and collaboration, problem solving, critical thinking, time management, adaptability, analytical thinking, attention to detail.

Domain Skills: Data analysis, CRISPR technology, bioremediation, biofertilizers, responsive web design, scripting in BioPython**.**

Dry Laboratory Skills: Data analysis, bioinformatics tools, in-silico gene and protein analysis, molecular docking, sequence alignment, statistical modelling.

Wet Laboratory Skills: Media preparation, microbial culture handling, PCR/RT-PCR, gel electrophoresis, blotting techniques, spectrophotometry, water and product testing, environmental monitoring, aseptic techniques, use of Laminar Air Flow.

Education

Lovely Professional University **2023-2027**
Bachelor's in technology- Biotechnology- CGPA: 7.2 *Phagwara, Punjab*

Tender Heart Senior Secondary School **2022 – 2023**
12th — Percentage: 86% *Ranchi, Jharkhand*

Tender Heart Senior Secondary School **2020 – 2021**
10th — CGPA: 7 *Ranchi, Jharkhand*